Customer Churn Prediction

1. Objective

The objective of this project is to build a supervised machine learning model to predict whether a customer will churn (leave the service) or not. This helps businesses identify potential churners and take preventive actions to improve customer retention.

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2. Dataset

- **Source:** Kaggle (Telco Customer Churn dataset)
- The dataset contains customer information such as demographics, account details, and service usage.
- Typical churn datasets include attributes like age, tenure, balance, and service usage patterns
- **Number of features:** 20
- **Target variable:** Churn (Yes/No)

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3. Methods

Data Preprocessing

- Removed missing/null values
- Encoded categorical variables
- Scaled numerical features using StandardScaler
- Performed train-test split (80% training, 20% testing)
- (If used) Applied SMOTE for class imbalance

Models Used

- Logistic Regression
- Random Forest Classifier

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4. Results

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Performance Summary

LR:

Accuracy= `0.917910447761194`

F1=`0.56`

ROC=

`0.7792717086834735`

RF:

Accuracy =`0.9626865671641791`

F1 =`0.8148148148148148`

ROC = `0.9106442577030812`

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5. Conclusion

The Random Forest model performed better than Logistic Regression across most evaluation metrics, particularly in terms of F1 Score and ROC-AUC.

This indicates that Random Forest is more effective in capturing complex patterns and identifying customers who are likely to churn.

Therefore, **Random Forest is the preferred model** for this task.

⭐ Advanced Add-on (Keep This — Important for Marks)

- Cross-validation ensured model reliability
- ROC curve analysis confirmed model performance
- Feature importance helped identify key churn drivers